

ADT Array

1)

The screenshot shows the SIMATS Online Programming Lab interface. The top header includes the SIMATS ENGINEERING logo, the title 'SIMATS Online Programming Lab', the subtitle 'DS PROGRAMMING', and a user profile for 'Rommala Varshini'. The left sidebar shows 'QUESTIONS 10 / 10 completed' and a dropdown menu for 'Integer Array'. The main area is titled 'PROGRAM EDITOR' and contains a C program for an integer array. The right sidebar shows the 'INPUT' field with the value '4'.

Integer Array

Sample Input:
Enter number of elements: 4
Enter elements:
10 20 30 40

Sample Output:
Using Array Indexing:
arr[0] = 10
arr[1] = 20
arr[2] = 30
arr[3] = 40

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i;
5     int *ptr;
6
7     printf("Enter number of elements: ");
8     scanf("%d", &n);
9
10    int arr[n];
11
12    printf("Enter elements:\n");
13    for (i = 0; i < n; i++) {
14        scanf("%d", &arr[i]);
15    }
16
17    ptr = arr;
18
19    printf("\nUsing Array Indexing:\n");
20    for (i = 0; i < n; i++) {
21        printf("arr[%d] = %d\n", i, arr[i]);
22    }
23 }
```

INPUT
4
10 20 30 40

2)

The screenshot shows the SIMATS Online Programming Lab interface. The top header includes the SIMATS ENGINEERING logo, the title 'SIMATS Online Programming Lab', the subtitle 'DS PROGRAMMING', and a user profile for 'Rommala Varshini'. The left sidebar shows 'QUESTIONS 10 / 10 completed' and a dropdown menu for 'Factorial'. The main area is titled 'PROGRAM EDITOR' and contains a C program for calculating the factorial of a number using recursion. The right sidebar shows the 'INPUT' field with the value '5'.

Factorial

Write a C program to find the factorial of a given number using recursion.

Sample Input: Enter a number- 6

Sample Output: Factorial of 6 is 720

```
1 // Factorial using recursion
2 #include<stdio.h>
3 long long fact(int n)
4 {
5     if(n==0|| n==1)
6         return 1;
7     if (n>1)
8         return n*fact(n-1);
9 }
10 int main()
11 {
12     int num ,i;
13     scanf("%d",&num);
14     if(num<0)
15         printf("fact cannot be defined for negative numbers");
16     else
17         printf("%d",fact(num));
18     return 0;
19 }
```

INPUT
5

3)

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QUESTIONS 10 / 10 completed
 Array Rotation
ARRAY ROTATION
 "Implement array rotation (left/right) for arrays of size 10 by K positions without using extra space. Use a reversal algorithm. The program must validate K ($0 \leq K < N$), show original→rotated→reversal steps visually, compute the sum

PROGRAM EDITOR C Run Save


```

1 #include<stdio.h>
2 void rev(int a[],int s,int e)
3 {
4     while(s<e)
5     {
6         int t=a[s];
7         a[s++]=a[e];
8         a[e--]=t;
9     }
10 }
11 int main()
12 {
13     int a[10],k,i;
14     char c;
15     for(i=0;i<10;i++)
16         scanf("%d",&a[i]);
17     scanf("%c",&c,&k);
18     if(c=='L' || c=='R')
19     {
20         rev(a,0,k-1);
21         rev(a,k,9);
        
```

INPUT
 1 2 3 4 5 6 7 8 9 0
 L
 3

4)

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QUESTIONS 10 / 10 completed
 Sparse Matrices
SPARSE MATRICES
 zero terms for two matrices. Add them, store result in sparse form (sorted by row then col), and display dense matrix form. Max 10x10; ignore zeros. Sample Input (Matrix 1):
 3 3 3
 0 1 5

PROGRAM EDITOR C Run Save


```

1 #include <stdio.h>
2 #define MAX 100
3 #define SIZE 10
4 typedef struct {
5     int row, col, value;
6 } Triple;
7 int main() {
8     Triple A[MAX], B[MAX], C[MAX];
9     int rows, cols, n1, n2;
10    int i, j, k = 0;
11    printf("Enter number of rows and columns (max 10x10): ");
12    scanf("%d %d", &rows, &cols);
13    printf("Enter number of non-zero elements in Matrix A: ");
14    scanf("%d", &n1);
15    printf("Enter row column value for Matrix A:\n");
16    for (i = 0; i < n1; i++) {
17        scanf("%d %d %d", &A[i].row, &A[i].col, &A[i].value);
18    }
19    printf("Enter number of non-zero elements in Matrix B: ");
20    scanf("%d", &n2);
21
        
```

INPUT
 Enter number of rows and columns (max 10x10): 3 3
 Enter number of non-zero elements in Matrix A: 2
 0 0 5
 1 2 8
 Enter number of non-zero elements in Matrix B: 2
 0 0 3
 2 1 6

5)

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QUESTIONS 10 / 10 completed

Fibonacci Series

FIBONACCI SERIES

Write a C program to print the first n terms of the Fibonacci series ($n \leq 20$) without using recursion. Start with 0 and 1.
Sample Input/Output:
Enter number of terms: 8
Fibonacci series: 0 1 1 2 3 5 8 13

PROGRAM EDITOR

C

Run Save

```

1 #include<stdio.h>
2 int main()
3 {
4     int n,a=0,b=1,c,i;
5     scanf("%d",&n);
6     for(i=1;i<=n;i++)
7     {
8         printf("%d ",a);
9         c=a+b;
10        a=b;
11        b=c;
12    }
13    return 0;
14 }
```

INPUT

4

33°C Light rain

Search

ENG IN

14:39 24-08-2026

6)

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QUESTIONS 10 / 10 completed

Array

ARRAY

Write a C program to find the 5th element in the given array/
Sample Input: a=[10,20,30,40,50,60]
Sample output: 50

PROGRAM EDITOR

C

Run Save

```

1 #include<stdio.h>
2 int main()
3 {
4     int a[100],n,i;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8     if(n>=5)
9         printf("%d",a[4]);
10    else
11        printf("not enough elements");
12    return 0;
13 }
```

INPUT

6
1 2 4 5 6 7

7)



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QUESTIONS 10 / 10 completed

Fibonacci series (N)

FIBONACCI SERIES (N ≤ 50)

Develop a C program to print the first N terms of the Fibonacci series (N ≤ 50) without using recursion. Use two variables to generate the series iteratively and store the series in an array. The program should also calculate and display the sum of

PROGRAM EDITOR

C
Run Save

```

1 #include<stdio.h>
2 int main()
3 {
4     int n,a=0,b=1,c,i,sum=0,f[50];
5     scanf("%d",&n);
6     if(n<1||n>50)
7     {
8         printf("invalid");
9         return 0;
10    }
11    for(i=0;i<n;i++)
12    {
13        f[i]=a;
14        printf("%d ",a);
15        sum+=a;
16        c=a+b;
17        a=b;
18        b=c;
19    }
20    printf("sum of series:%d\n",sum);
21    printf("largest term:%d\n",f[n-1]);

```

OUTPUT

INPUT

10

8)



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QUESTIONS 10 / 10 completed

Unsorted Array

UNSORTED ARRAY

Write a program to find the missing element for the given unsorted array of sequence numbers.
Sample Input: a=[10,9,7,4,3,1,2]
Sample output: 5,6,8

PROGRAM EDITOR

C
Run Save

```

1 #include<stdio.h>
2 int main()
3 {
4     int a[100],n,i,j,max,f;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7     {
8         scanf("%d",&a[i]);
9         max=a[i];
10        for(i=1;i<n;i++)
11        {
12            if(a[i]>max)
13            {
14                max=a[i];
15            }
16        }
17        for(i=1;i<=max;i++)
18        {
19            f=0;
20            for(j=0;j<n;j++)
21            {
22                if(a[j]==i)
23                {
24                    f=1;
25                }
26            }
27            if(f==0)
28            {
29                printf("%d ",i);
30            }
31        }
32    }
33    return 0;

```

OUTPUT

INPUT

7
10 9 7 4 3 2 1

9)

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QUESTIONS 10 / 10 completed
 Fib series-iterative

FIB SERIES-ITERATIVE & RECUR
 Write a C program to generate Fibonacci series up to N terms using both iterative and recursive methods. The program should display both outputs and compare execution time in milliseconds. Also display total recursive calls made.

PROGRAM EDITOR C Run Save


```

1 #include<stdio.h>
2 int fib(int n)
3 {
4     if(n<=1)
5         return n;
6     return fib(n-1)+fib(n-2);
7 }
8 int main()
9 {
10     int n,i,a=0,b=1,c;
11     scanf("%d",&n);
12     printf("iterative:\n");
13     for(i=0;i<n;i++)
14     {
15         printf("%d ",a);
16         c=a+b;
17         a=b;
18         b=c;
19     }
20     printf("\nrecursive:\n");
21     for(i=0;i<n;i++)
    
```

INPUT
 6

OUTPUT

10)

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QUESTIONS 10 / 10 completed
 Structure

STRUCTURE
 Write a C program to store student details (Roll No, Name, Marks) using structure and save them into a file. Display highest mark, class average, and total records stored.
 Sample Input:
 Enter number of

PROGRAM EDITOR C Run Save


```

1 #include<stdio.h>
2 struct student
3 {
4     int roll,marks;
5     char name[20];
6 };
7 int main()
8 {
9     struct student s[20];
10    FILE*fp;
11    int n,i,sum=0,max=0,pos=0;
12    fp=fopen("student.txt","w");
13    scanf("%d",&n);
14    for(i=0;i<n;i++)
15    {
16        scanf("%d%s%d",&s[i].roll,&s[i].name,&s[i].marks);
17        fprintf(fp,"%d %s %d\n",s[i].roll,s[i].name,s[i].marks);
18        sum+=s[i].marks;
19        if(s[i].marks>max)
20        {
21            max=s[i].marks;
    
```

INPUT
 2
 101 Ravi 75
 102 Kiran 85